

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX Variable Mag Adjustment Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	08 May 2015	First Release	CHIN CHEE YAN
01	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

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Caution, all steps MUST perform with Inner Barrier Open position.

Variable Mag Adjustment Steps:

1. Ensure the USB 2.0 A to Micro-B Cable 3M (P/N, 2560-0024) is connected to Mini-ITX Industrial Computer (Atom PC-vertical riser) (P/N,2160-0044) as shown.



Figure 1

2. At DIO tab, ensure input bit for Inner barrier input bit is **"Open"** status and the Z-Axis Servo On is **"Disable"**. If not, manually toggle output bit as shown.

Services → Subsystem Status and Control → DIO → Initialize → Inner Barrier to "Open" status → "Disable" Z-Axis Enable Signal On

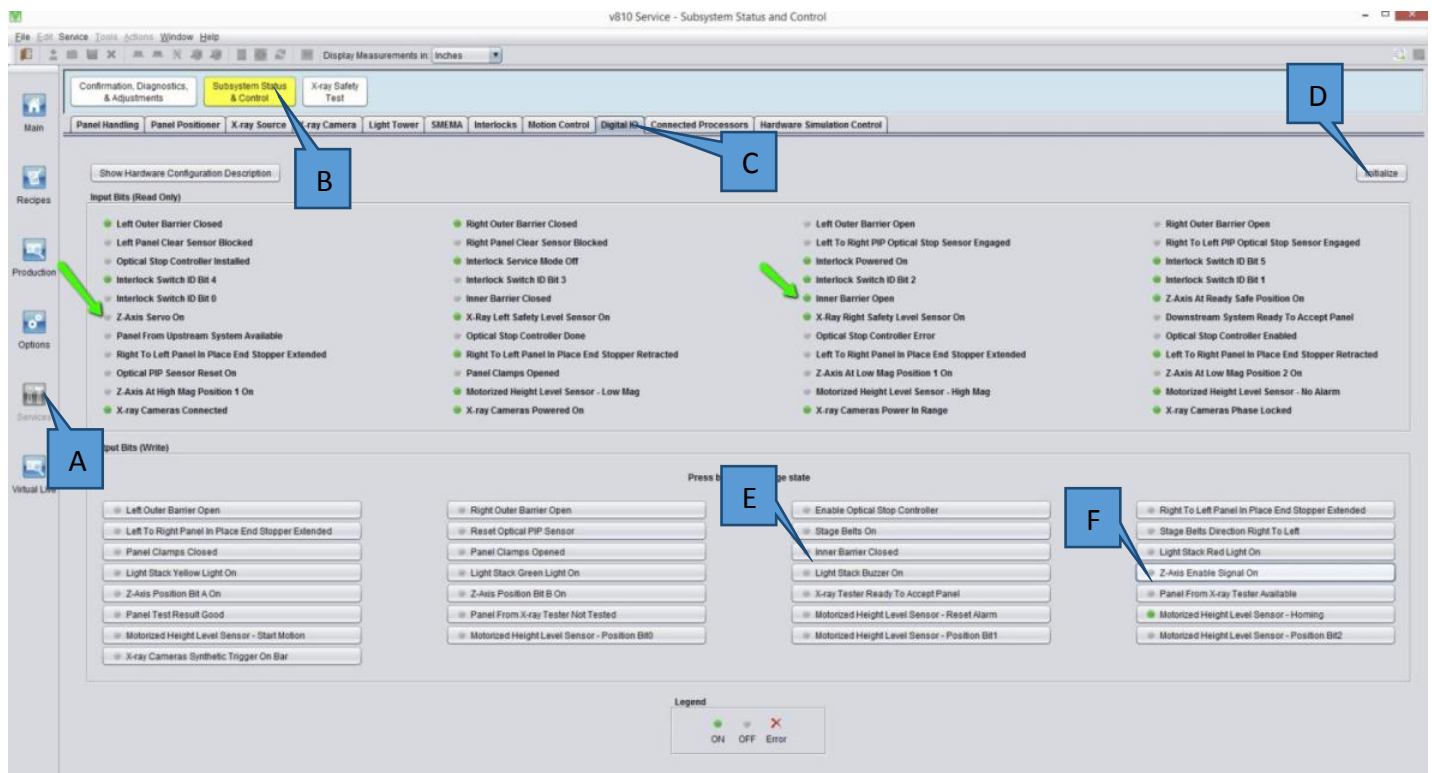


Figure 2

3. Remote desktop to into Atom PC with IP, **192.168.128.70**, User, **Administrator**, Password, **Please!**

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- At the Atom PC desktop screen, launch the “ClearPath MSP Software” as shown. Ensure the motor status is showing “Disable” at the bottom left corner of the application. If not back to DIO tab, disable the Z-Axis Servo On.

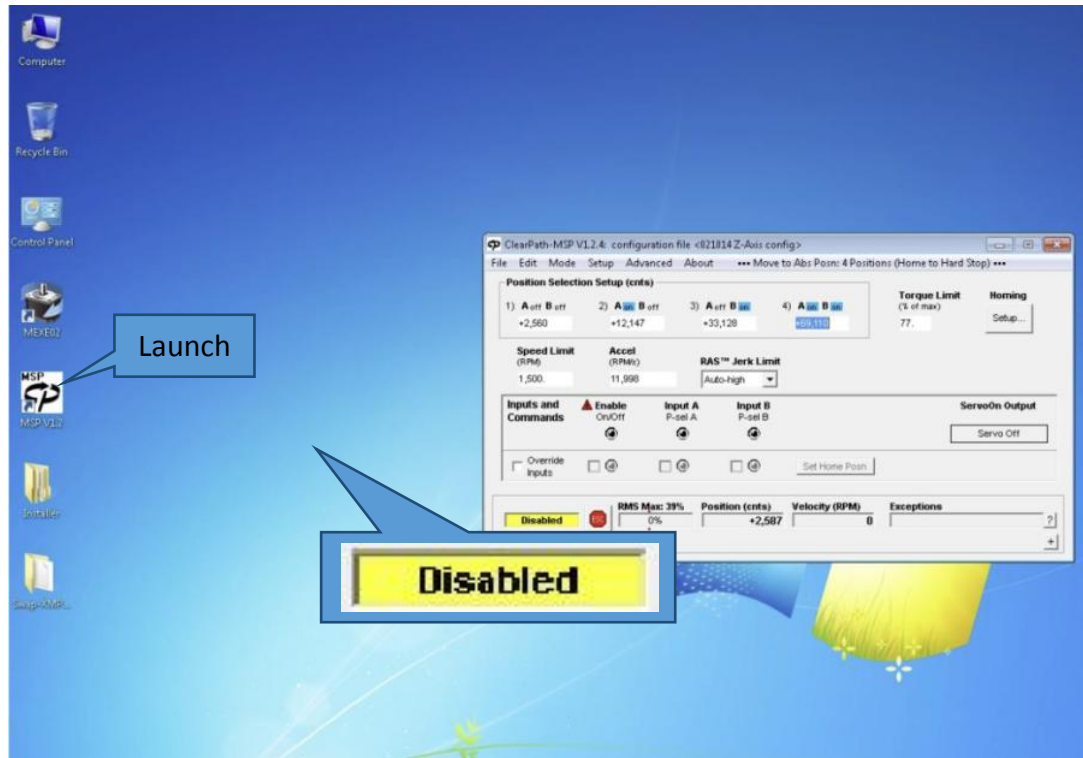


Figure 3

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- Go to path “C:\Users\Administrator\Desktop\Motor Config Files\Z-Axis” and back-up the origin folder. #Should be containing 1 file as default manufacture configure.

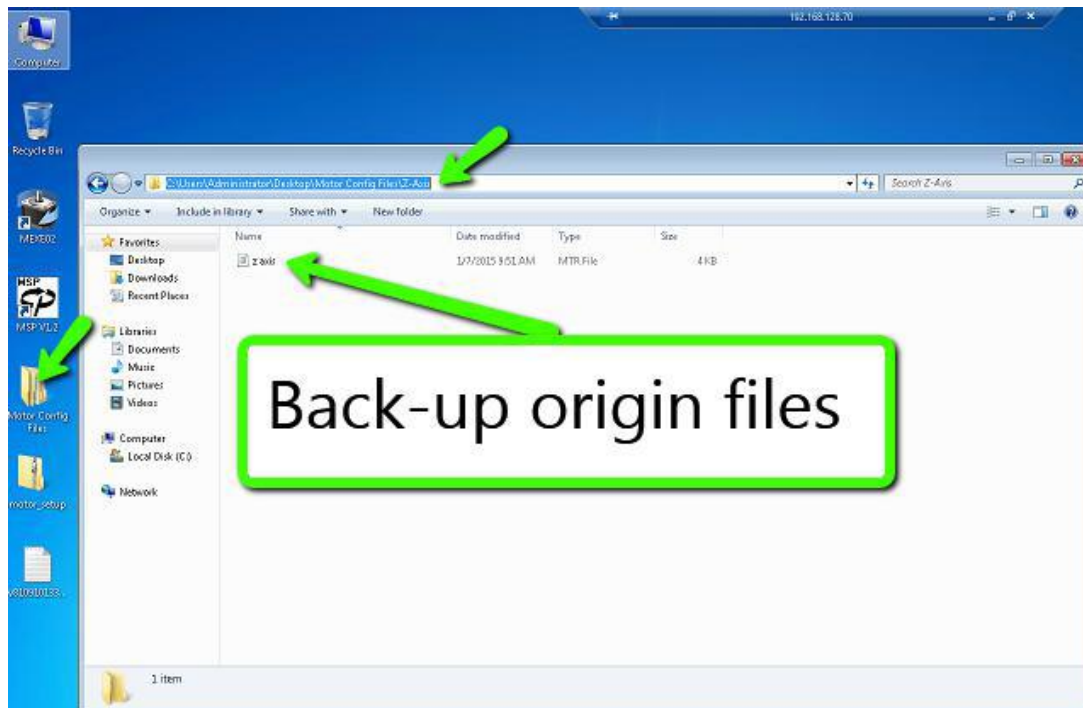


Figure 4

- Browses to path “C:\Users\Administrator\Desktop\Motor Config Files\Z-Axis” and load the existing Z motor config. Should be containing 1 file as default manufacture configure.

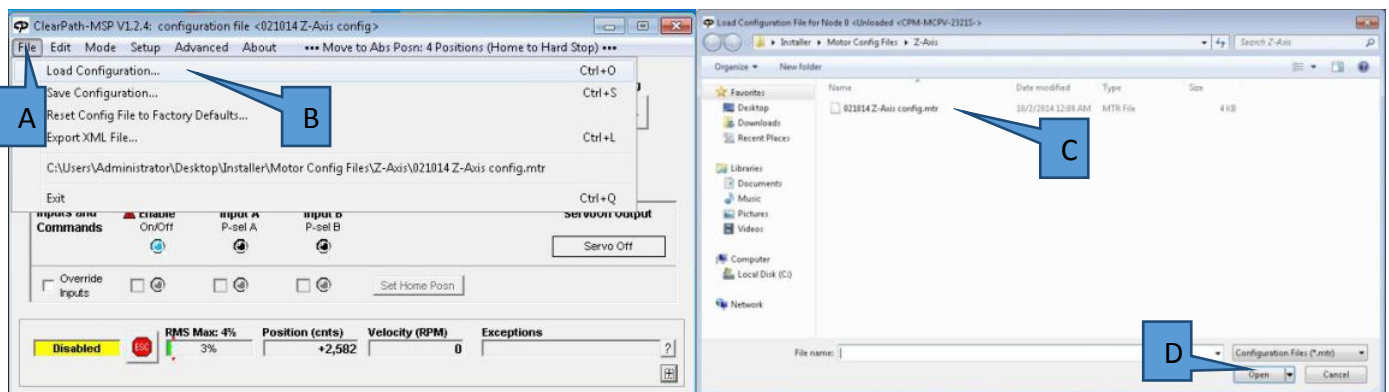


Figure 5

7. Once configuration loading completed, Z height threshold will automatic updated as shown.

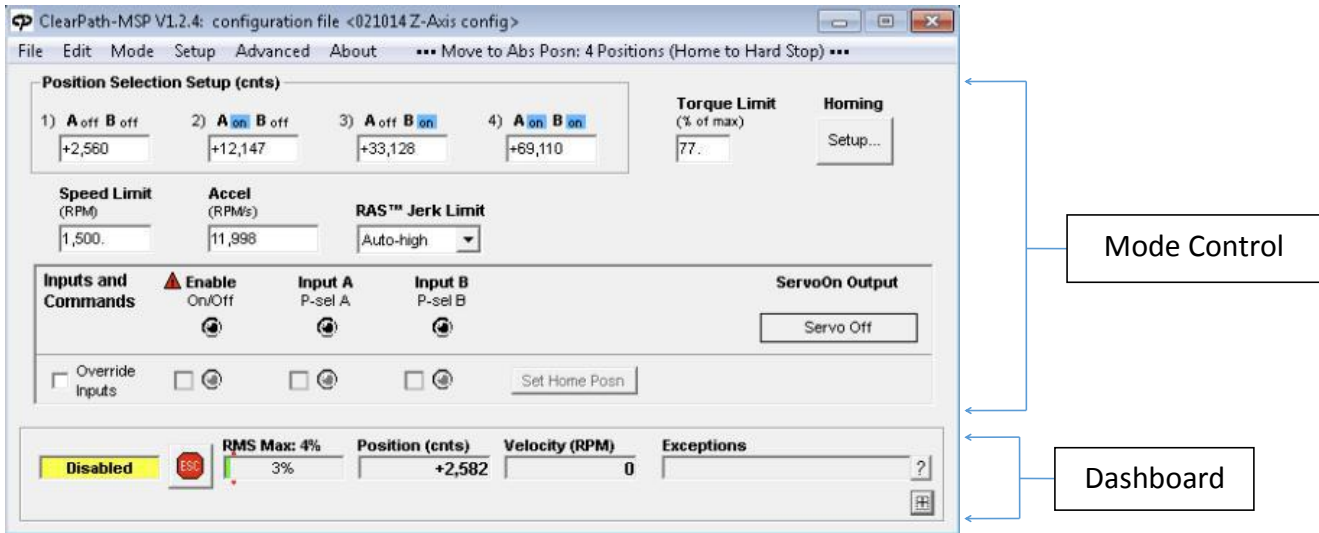


Figure 6

8. Check on the “Override Inputs” and check “Enable” to allow MSP application to drive Z motor. Once completed, motor status will display as “SW Enable”.

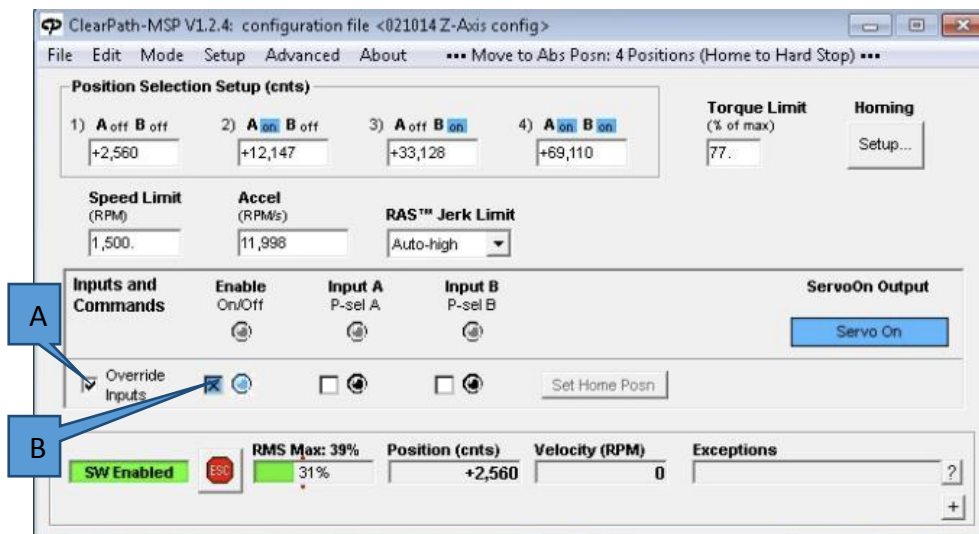


Figure 7

9. Once completed, back to DIO I/O tab page, check on the “Input Bit”, ensure the “Z-Axis at Safe Position On” turn to Green. If the indicator does not turns to Green (Grey), need re-position the Z-Axis positioning sensor.

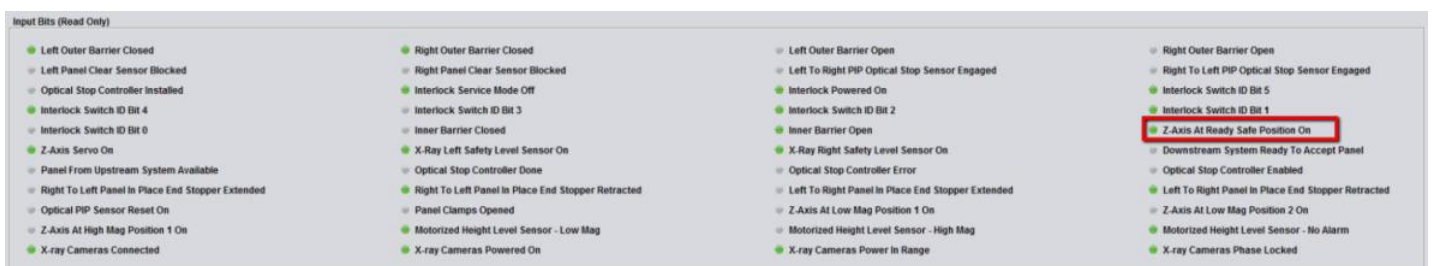


Figure 8

10. Assume sensor need to re-position, turn off X-Ray source and Open the Front Access Door. Used

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temper Proof kit unscrew and unscrew and open Right Outer Barrier.

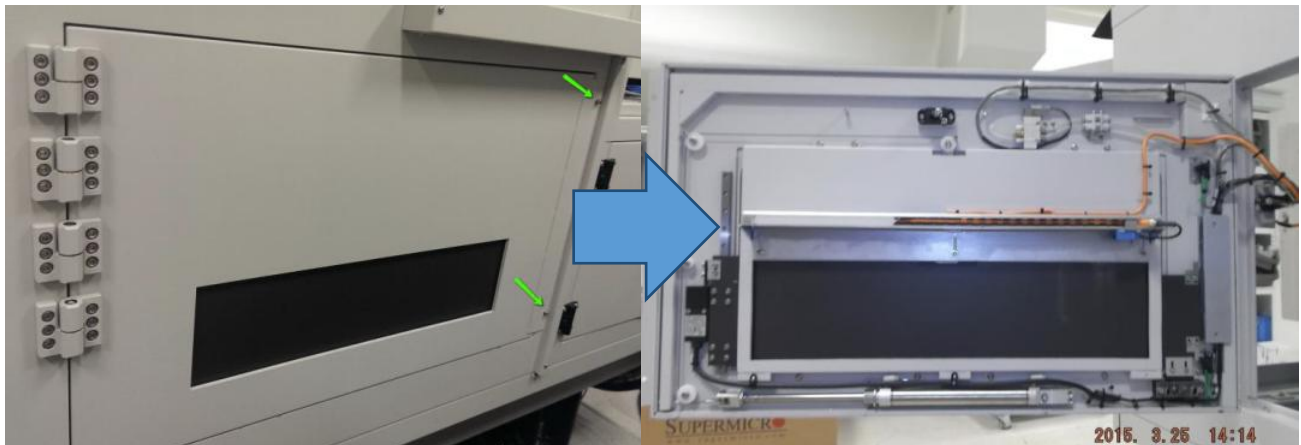


Figure 9

11. Access from right outer barrier, unscrew (M7 nut x 6) as pointed and remove the U-COLUMN WINDOW ASSEMBLY (P/N, 4JAHW-A049) as shown.

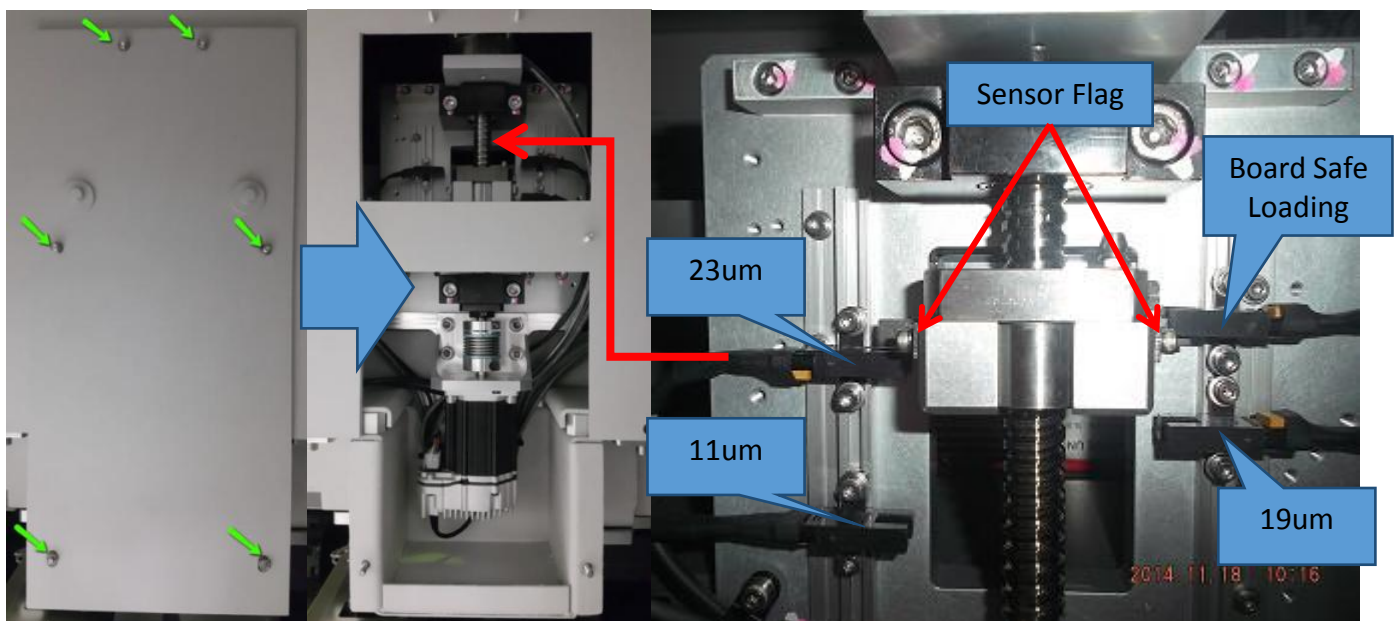


Figure 10

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12. Reposition the board safe loading sensor with loosens screws as shown. Move the sensor toward to bottom until the Red Indicator light ON (camera flag should cover by sensor) and tighten the screws. Back to DIO tab and ensure the “Z-Axis at Safe Position On” turn to **Green**.

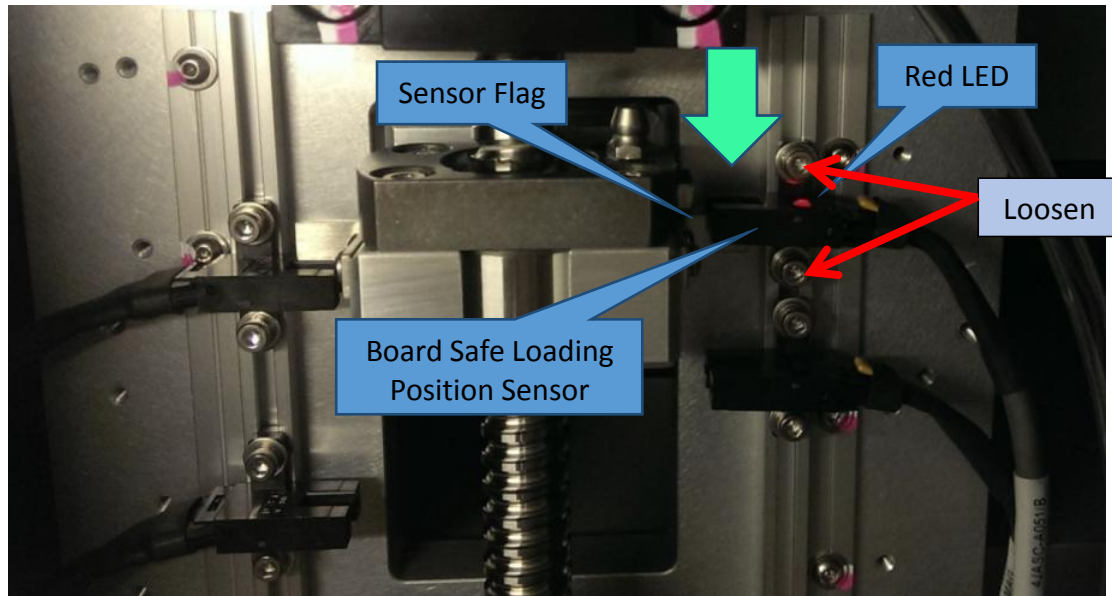


Figure 11

13. Back to MSP application, check on the “Input A” and tube will go to 23um position.

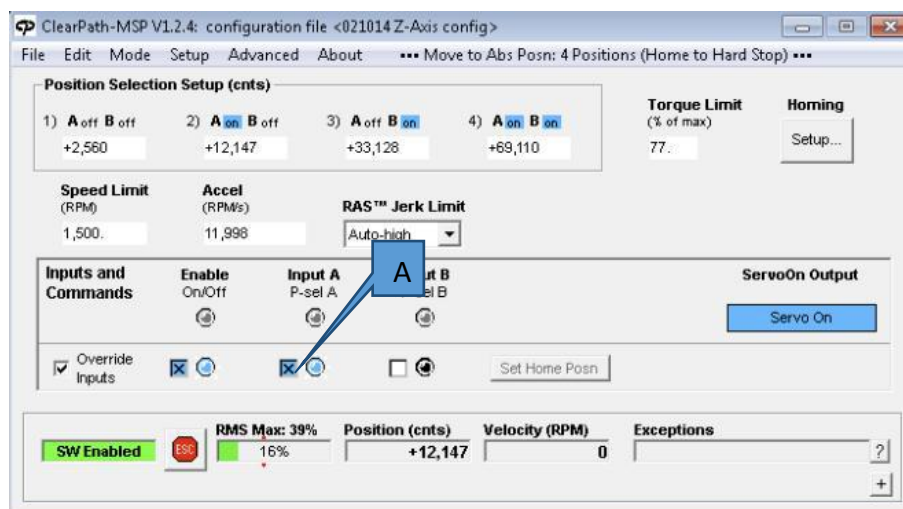


Figure 12

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13. Once completed, back to DIO I/O tab page, ensure the “Z-Axis at Low Mag Position 1 On” turn to Green. If the indicator does not turns to Green (Grey), need re-position the Z-Axis positioning sensor.



Figure 13

14. Reposition the board safe loading sensor with loosens screws as shown. Move the sensor toward to bottom until the Red Indicator light ON (camera flag should cover by sensor) and tighten the screws. Back to DIO tab and ensure the “Z-Axis at Low Mag Position 1 On” turn to **Green**.

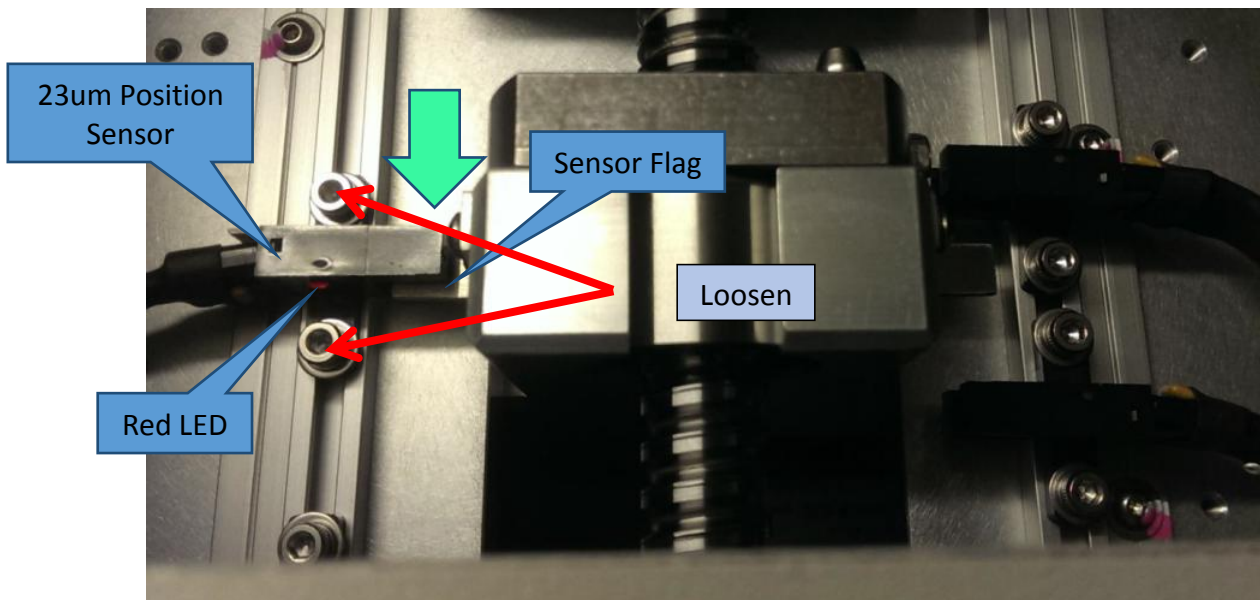


Figure 14

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14. Back to MSP application, uncheck on the “Input A” and check “Input B”, tube will go to 19um position.

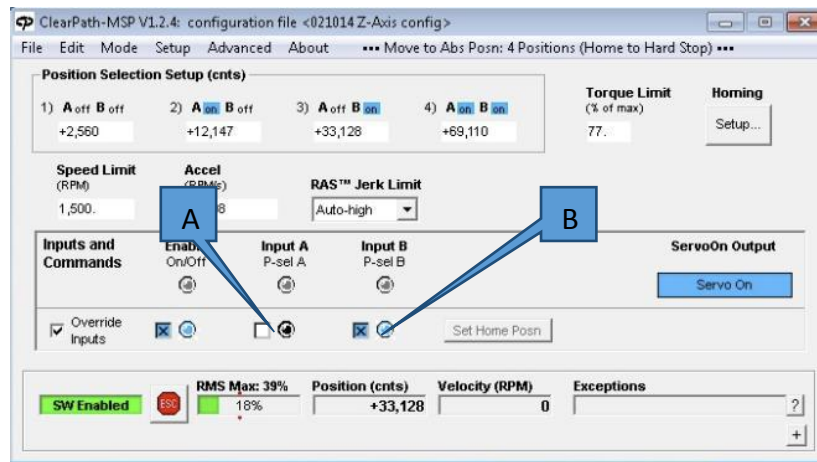


Figure 15

15. Once completed, back to DIO I/O tab page, ensure the “Z-Axis at Low Mag Position 2 On” turn to Green. If the indicator does not turn to Green (Grey), need re-position the Z-Axis positioning sensor.



Figure 16

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16. Reposition the board safe loading sensor with loosens screws as shown. Move the sensor toward to bottom until the Red Indicator light ON (camera flag should cover by sensor) and tighten the screws. Back to DIO tab and ensure the “Z-Axis at Low Mag Position 2 On” turn to **Green**.

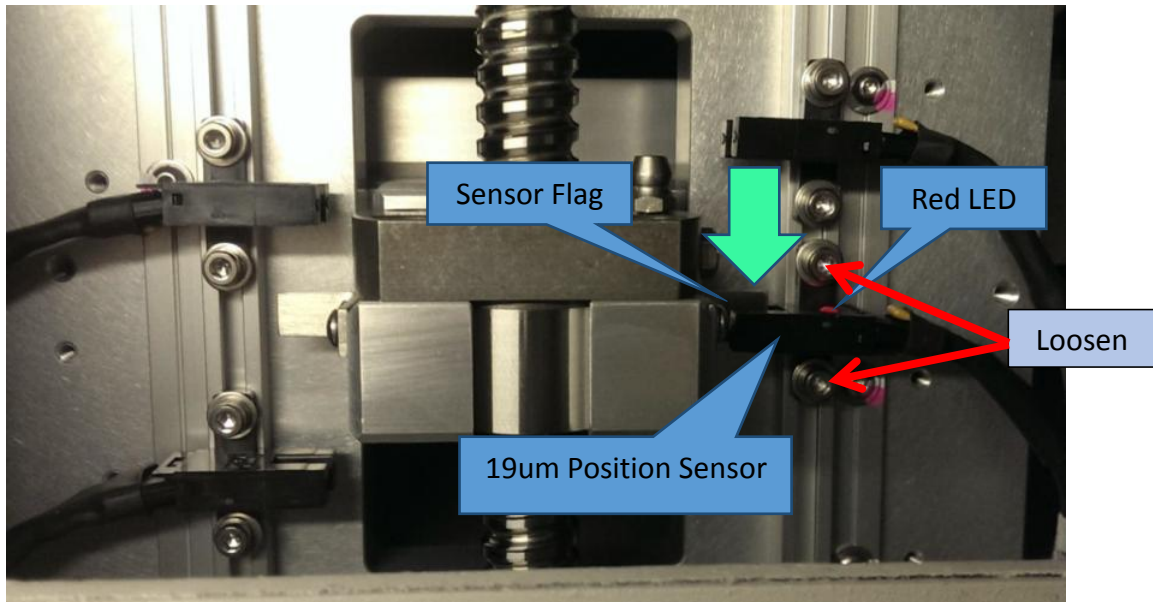


Figure 17

15. Back to MSP application, check the “Input A” and “Input B”, tube will go to 11um position.

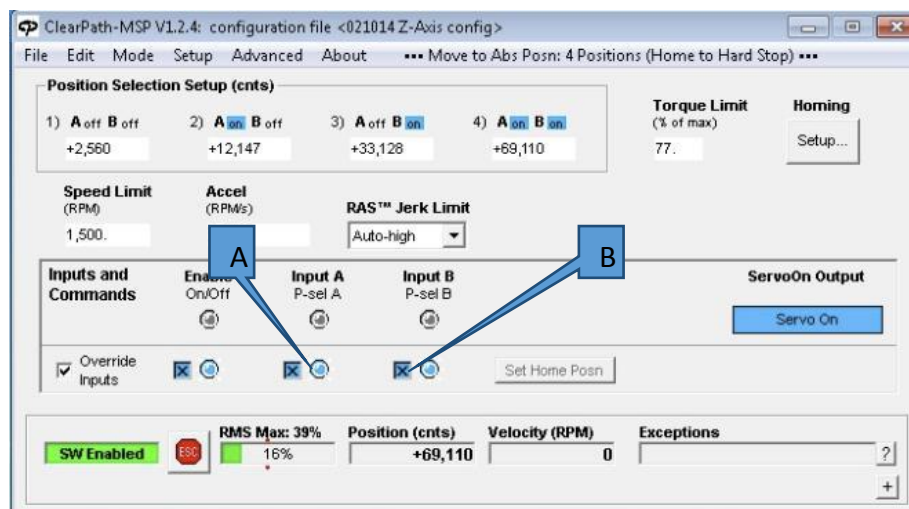


Figure 18

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17. Once completed, back to DIO I/O tab page, ensure the “Z-Axis at High Mag Position 1 On” turn to Green. If the indicator does not turns to Green (Grey), need re-position the Z-Axis positioning sensor.



Figure 19

18. Reposition the board safe loading sensor with loosens screws as shown. Move the sensor toward to bottom until the Red Indicator light ON (camera flag should cover by sensor) and tighten the screws. Back to DIO tab and ensure the “Z-Axis at High Mag Position 1 On” turn to **Green**.

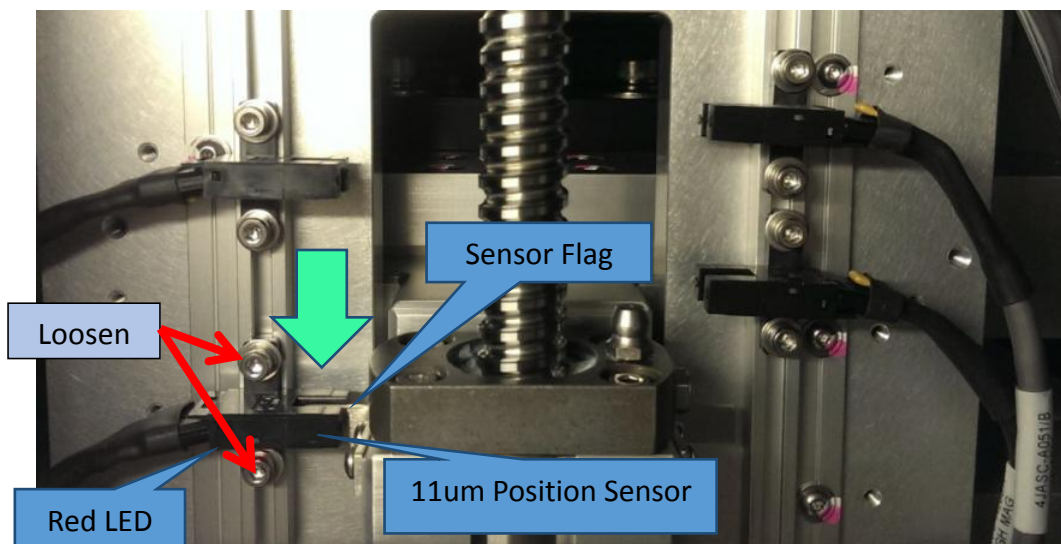


Figure 20

19. Once completed, uncheck all checked item.

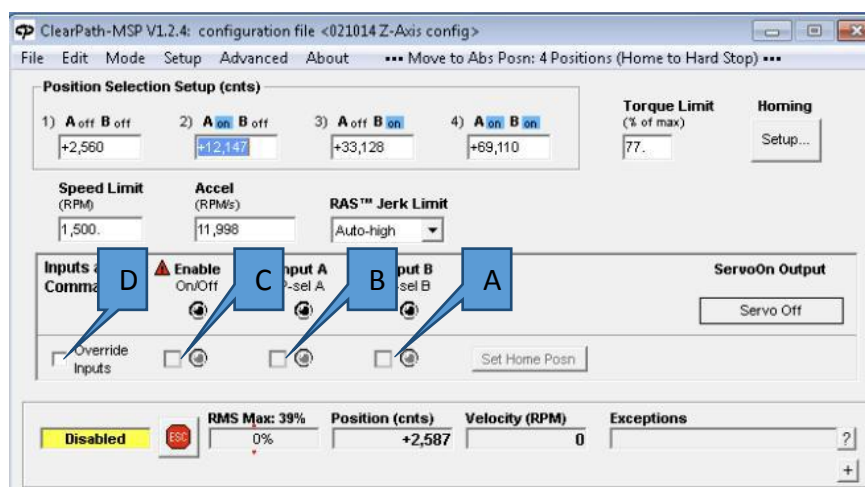


Figure 21

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20. Back to DIO tab, enable the “Z-Axis Servo On” at Output Bits. Check at Input bit “Z-Axis Servo On” and “Z-Axis Ready Safe Position On” should show Green.



Figure 22

21. Remove all tools and others extra material from machine, close and lock the right outer barrier and front access doors.

22. Check the current magnification resolutions in used at v810 GUI main tab. (eg. 19um and 11um).

Change the setting for Low Mag to 23um. Close and restart GUI once setting changes.

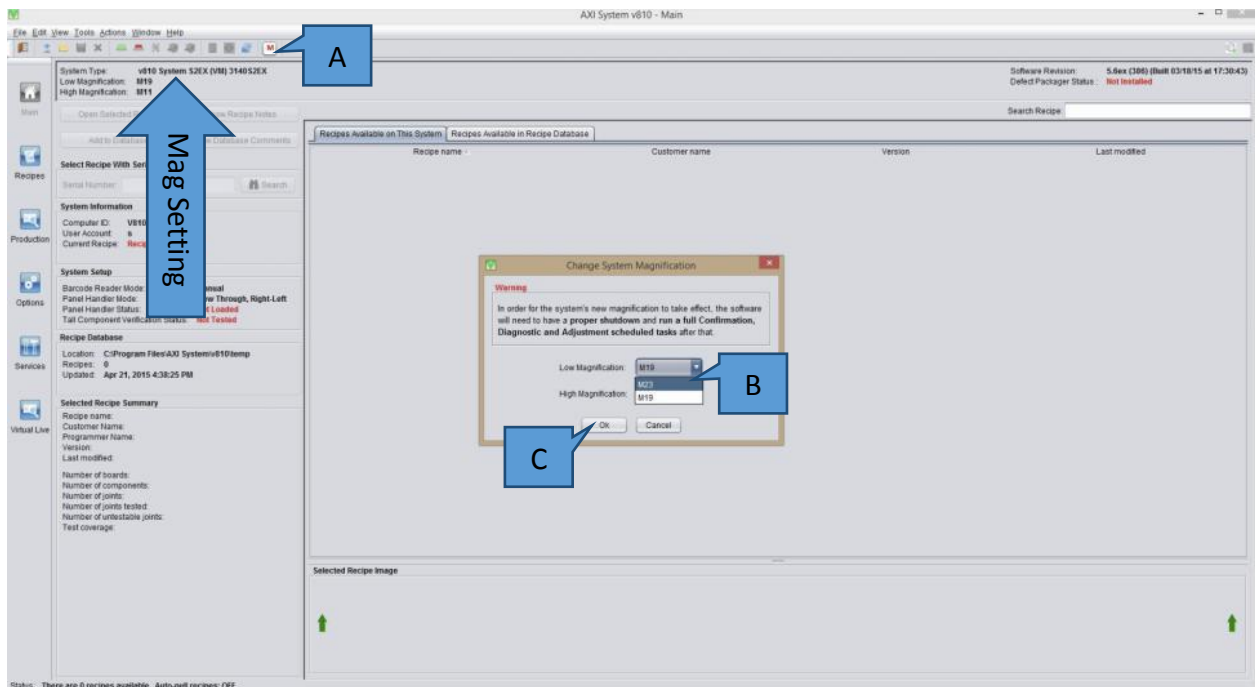


Figure 23

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20. Go to Confirmation, Diagnostics, & Adjustment tab, right on the schedule task and select [Loop Once] to run the full loop of CD&A with default motor counts settings.

Services → Confirmation, Diagnostic and Adjustment → Scheduled Tasks → Run 1 loop

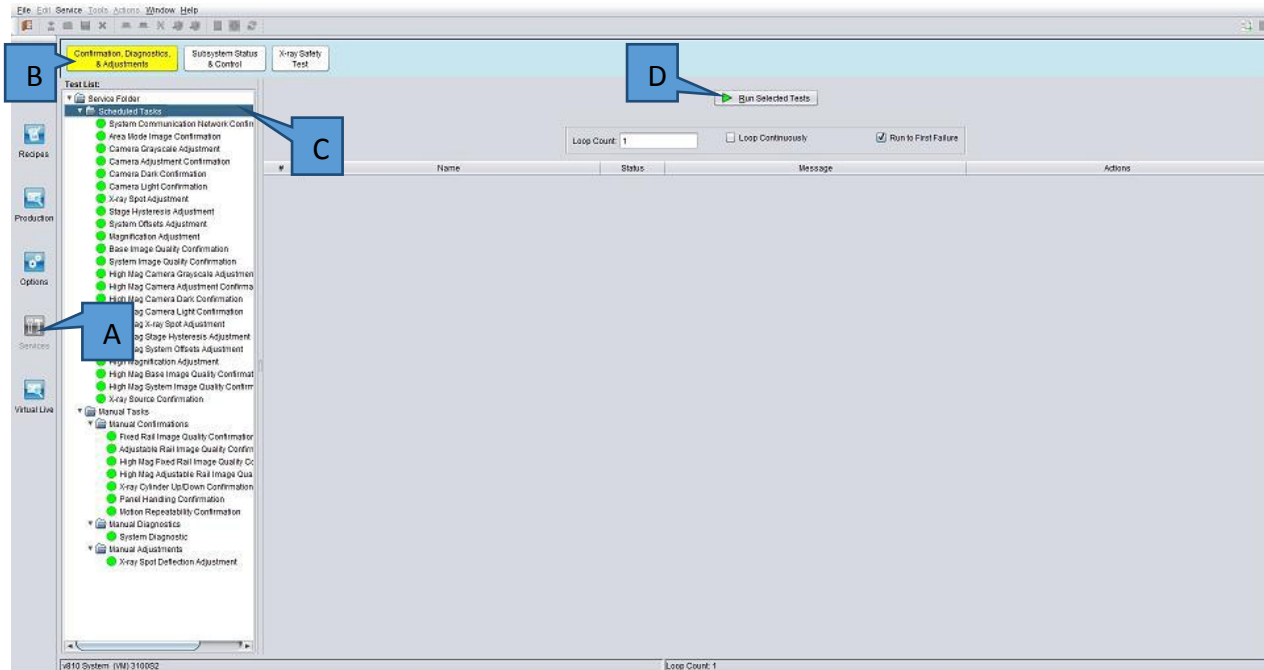


Figure 24

21. Go the path "C:\Program Files\AXI System\v810\<Rev>\log\calib\Magnification". Open and check the latest low magnification factor at the updated log files value with sort by modify date.

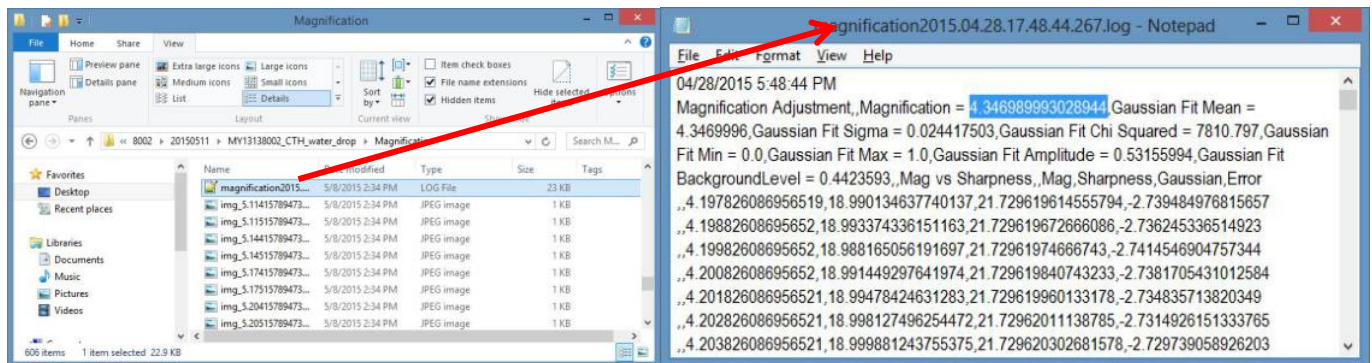


Figure 25

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22. Magnification spec for 23um is within range 4.33 ~ 4.35. Base on log files value, perform a calculation with formulae below to calculate the motor count modify result.

$$\underline{7100506.11 (\delta M)}$$

$$249.17 - 74.48 (\delta M)$$

δM = Magnification spec - Current Magnification Value

Exampel;

- 23um Magnification Spec = **4.34**
- Current 23um Magnification Value = 4.28

Calculation;

$$7100506.11(4.34 - 4.28) / 249.17 - 74.48(4.34 - 4.28) = \mathbf{1741.023}$$

23.Go to digital I/O tab to disable the [Z-axis Enable Signal On].

24.Back to Atom PC, change the motor counts for **AonBoff** according to the 23um formulae. Must click “Enter” to update the motor count value into system.

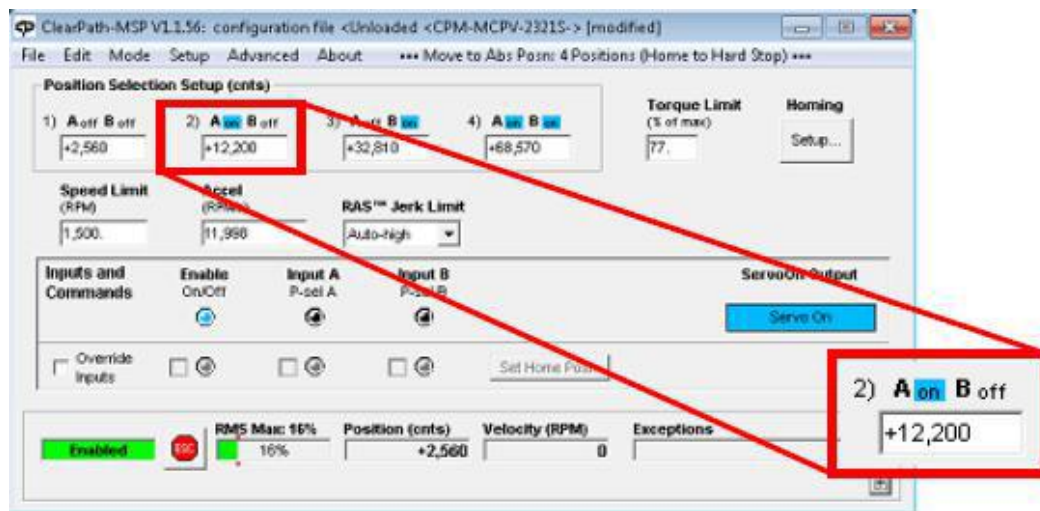


Figure 26

Remark,

- Higher value, physical tube become lower.
- If formulae result is positive, add into existing value, if negative, minus the existing value.

25. After modify the motor counts, back to DIO tab and enable the “Z-axis Enable Signal On” and rerun CD&A as step 20 ~ 21. Check the update magnification value. Do some minor counts adjustment by repeating step 25 until magnification factor within range 4.33 ~ 4.35.

Minor change = +/-100 or 50 count

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26. Once pass 23um Low Mag adjustment, continues with 11um High Mag optimization. Go the path “C:\Program Files\AXI System\v810\<Rev>\log\calib\Magnification”. Open and check the latest high magnification factor at the updated log files value with sort by modify date.

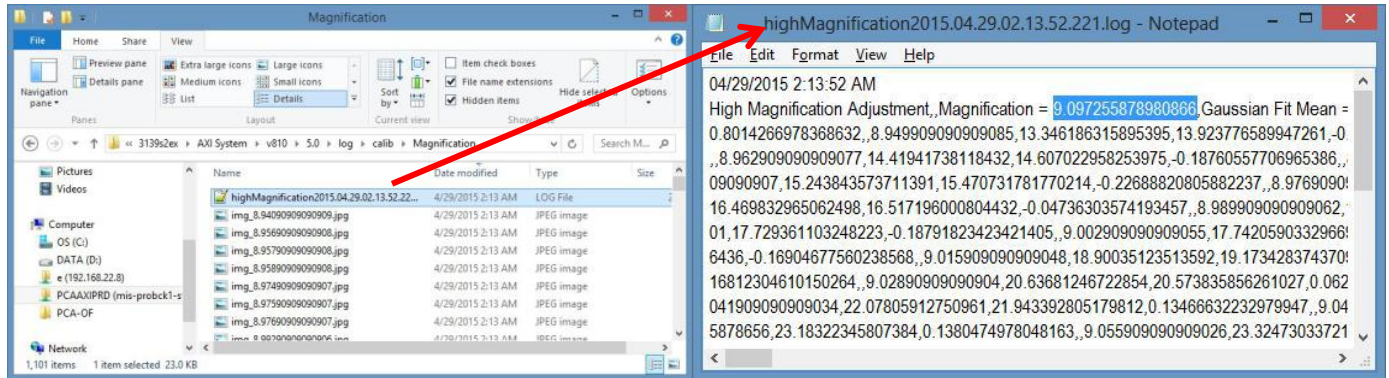


Figure 27

27. Magnification spec for 11um is within range 9.07 ~ 9.09. Base on log files value, perform a calculation with formulae below to calculate the motor count modify result.

$$\underline{1213470.85 \delta M}$$

$$249.17 - 30.79 \delta M$$

δM = Magnification spec - Current Magnification Value

Exampel;

- 11um Magnification Spec = **9.09**
- Current 11um Magnification Value = 9.16

Calculation;

$$1213470.85 (9.09-9.16)/ 249.17 - 30.79(9.09-9.16) = \mathbf{-337.99}$$

28. Go to digital I/O tab to disable the [Z-axis Enable Signal On].

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29. Back to Atom PC, change the motor counts for **A on B on** according to the 11um formulae. Must click “Enter” to update the motor count value into system.

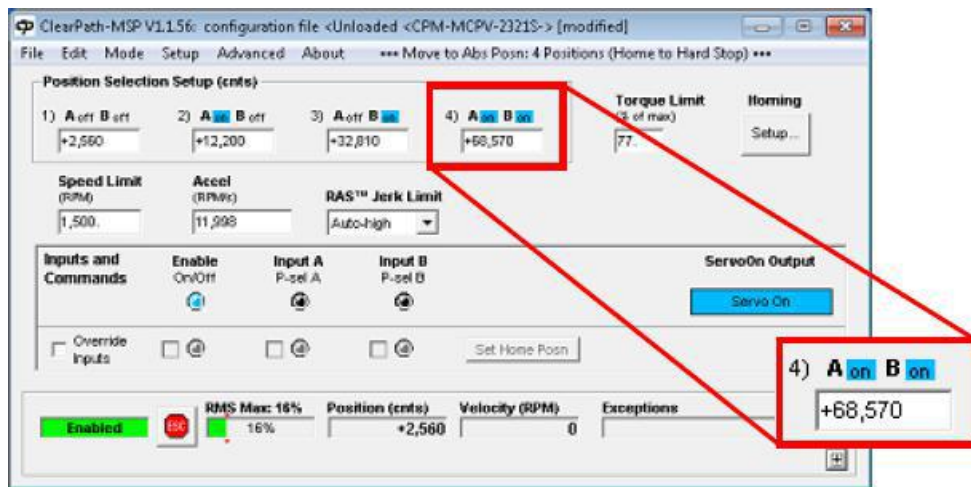


Figure 28

Remark,

- Higher value, physical tube become lower.
- If formulae result is positive, add into existing value, if negative, minus the existing value.

30. After modify the motor counts, back to DIO tab and enable the “Z-axis Enable Signal On” and rerun CD&A as step 20 ~ 21. Check the update magnification value. Do some minor counts adjustment by repeating step 29 until magnification factor within range 9.07 ~ 9.09.

Minor change = +/-100 or 50 count

31. After completed the 23um and 11um resolutions, change the low magnification setting to 19um at v810 GUI main tab by toggling magnification change button. Close and restart GUI once setting changes.

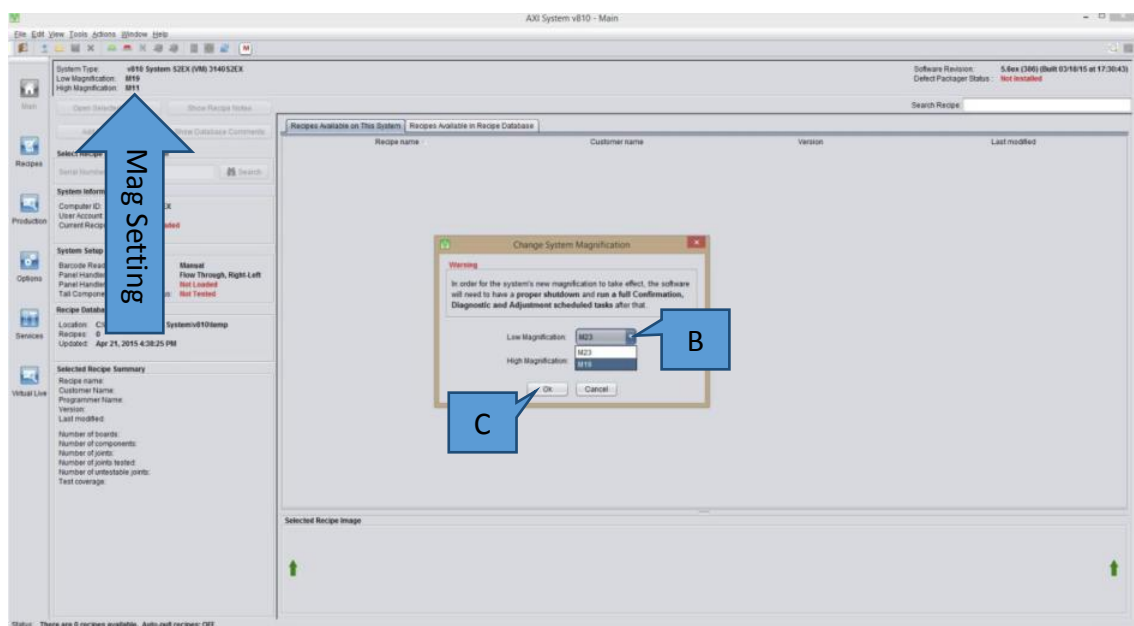


Figure 29

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32. Go to Confirmation, Diagnostics, & Adjustment tab, right on the schedule task and select [Loop Once] to run the full loop of CD&A with default motor counts settings.

Services → Confirmation, Diagnostic and Adjustment → Scheduled Tasks → Run 1 loop

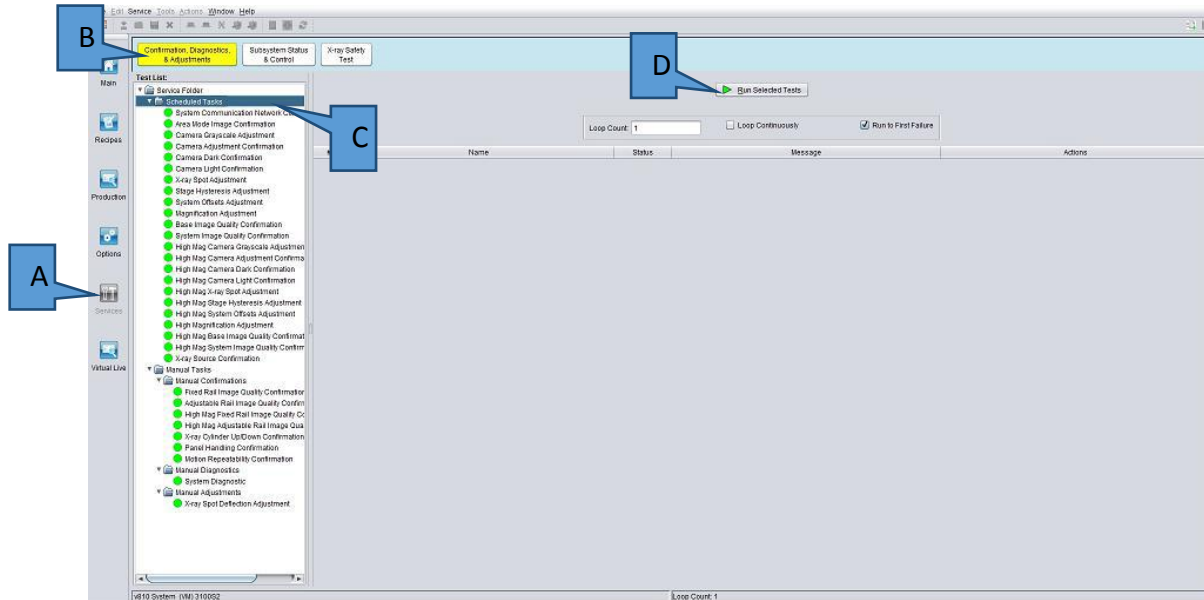


Figure 30

33. Go the path "C:\Program Files\AXI System\v810\<Rev>\log\calib\Magnification". Open and check the latest low magnification factor at the updated log files value with sort by modify date.

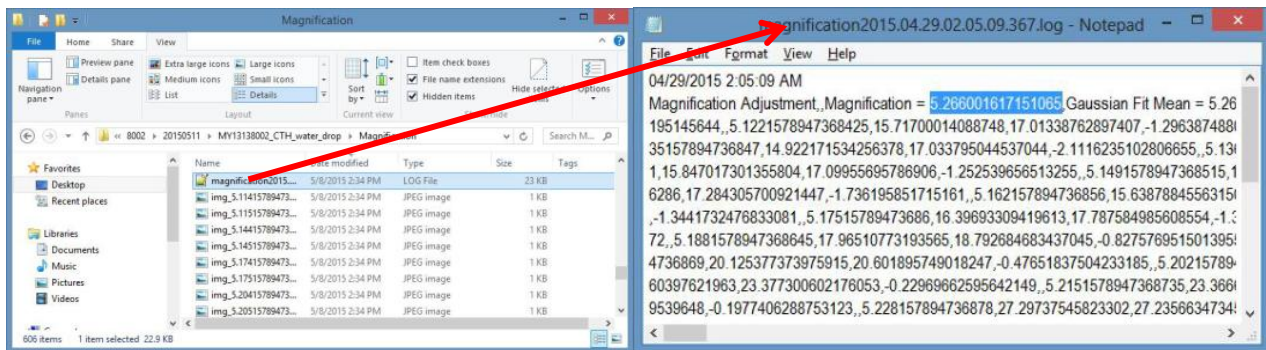


Figure 31

34. Magnification spec for 19um is within range 5.24 ~ 5.26. Base on log files value, perform a calculation with formulae below to calculate the motor count modify result.

$$\frac{4378982.53}{\delta M}$$

$$249.17 - 58.49 \delta M$$

$$\delta M = \text{Magnification spec} - \text{Current Magnification Value}$$

Example;

- 19um Magnification Spec = **5.26**
- Current 23um Magnification Value = 5.19

Calculation;

$$4378982.53(5.26 - 5.19) / 249.17 - 58.49 (5.26 - 5.19) = 1250.751$$

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35. Go to digital I/O tab to disable the [Z-axis Enable Signal On].

36. Back to Atom PC, change the motor counts for **A off B on** according to the 19um formulae. Must click **“Enter”** to update the motor count value into system.

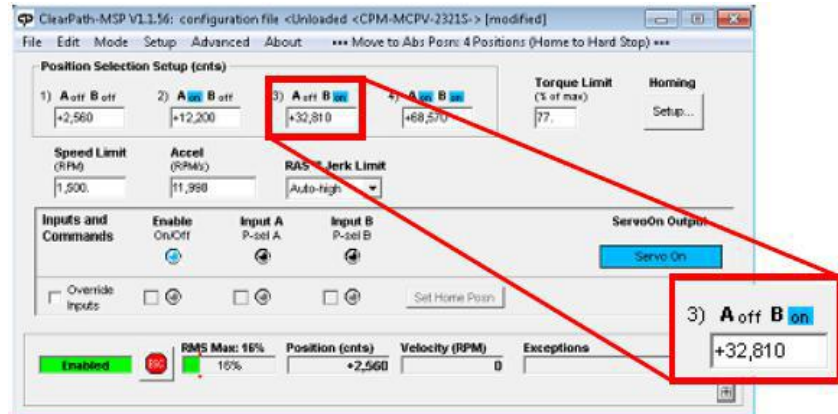


Figure 32

Remark,

- Higher value, physical tube become lower.
- If formulae result is positive, add into existing value, if negative, minus the existing value.

37. After modify the motor counts, back to DIO tab and enable the “Z-axis Enable Signal On” and rerun CD&A as step 32 ~ 33. Check the update magnification value. Do some minor counts adjustment by repeating step 36 until magnification factor within range 5.24 ~ 5.26.

Minor change = +/-100 or 50 count

38. After the adjustment completed for all magnification (11um, 19um, 23um), click [Save As] to save the fine-tuned ClearPath motor config file.

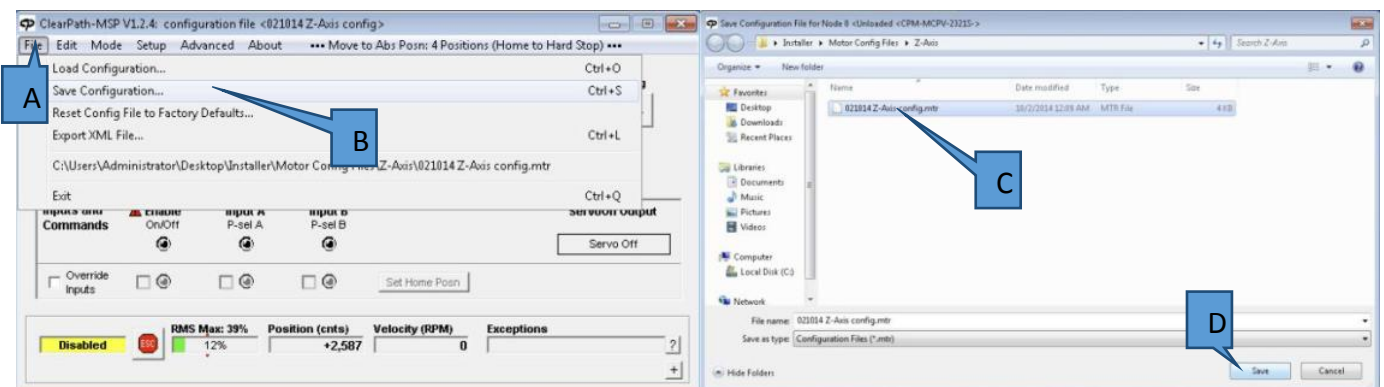


Figure 33

39. Back to DIO tab, manually toggle bit below and ensure the sensor is trigger consistently.

Position	I/O Bit
	<u>Safe Board Loading Position</u> Input ☐ Z-Axis Position Bit A On ☐ Z-Axis Position Bit B On Output ☒ Z-Axis At Ready Safe Position On
	<u>23um Resolution</u> Input ☒ Z-Axis Position Bit A On ☐ Z-Axis Position Bit B On Output ☒ Z-Axis At Low Mag Position 1 On
	<u>19um Resolution</u> Input ☐ Z-Axis Position Bit A On ☒ Z-Axis Position Bit B On Output ☒ Z-Axis At Low Mag Position 2 On
	<u>11um Resolution</u> Input ☒ Z-Axis Position Bit A On ☒ Z-Axis Position Bit B On Output ☒ Z-Axis At High Mag Position 1 On

Figure 34

40. Once completed, assembly the U-COLUMN WINDOW ASSEMBLY (P/N, 4JAHW-A049) back to origin.
Run CDnA for 5 loops (23um and 11um /19um and 11um) each and verify the magnification value accuracy for each mag.